

Natural language statement

Domain

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1.18 Corollary. If $f(e^{i\theta}) = \sum a_n e^{in\theta}$ with $\sum a_n < \infty$, and f never vanishes, then $1/f(e^{i\theta}) = \sum b_n e^{in\theta}$ with $\sum b_n < \infty$. Notation: the sums run over all n in $\mathbb{Z}$. The corollary is read off from Theorem 1.17, which identifies the Gelfand spectrum $\sigma(\mathcal{A})$ of the convolution algebra $\mathcal{A}^1(\mathbb{Z})$ with the unit circle $\mathbb{T}$ so that the Gelfand transform becomes $a(e^{i\theta}) = \sum_{-\infty}^{\infty} a_n e^{in\theta}$.	Harmonic analysis	theorem folland_1_18_wiener_inverse_of_absolutely_convergent_series (a : $\mathbb{Z} \rightarrow \mathbb{C}$) (ha : $\sum_{n : \mathbb{Z}} a_n < \infty$) (hne : $\forall \theta : \mathbb{R}, (\sum' n : \mathbb{Z}, a_n * \text{Complex.exp } (n * \theta * \text{Complex.I})) \neq 0$) : exists b : $\mathbb{Z} \rightarrow \mathbb{C}, (\sum_{n : \mathbb{Z}} b_n < \infty) \wedge \forall \theta : \mathbb{R}, (\sum' n : \mathbb{Z}, b_n * \text{Complex.exp } (n * \theta * \text{Complex.I})) * (\sum' n : \mathbb{Z}, a_n * \text{Complex.exp } (n * \theta * \text{Complex.I})) = 1$:= by sorry
2.29 Proposition. If $G/[G,G]$ is compact, then G is unimodular. Notation: $[G,G]$ denotes the smallest closed subgroup of G containing all elements of the form $[x,y] = xyx^{-1}y^{-1}$; it is called the commutator subgroup of G , and it is normal since $z[x,y]z^{-1} = [zxz^{-1}, zyz^{-1}]$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right translates of f , and $f^*g(x) = \int f(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. The modular function $\Delta : G \rightarrow (0,\infty)$ is determined by $\lambda(\Delta x) = \Delta(x)\lambda(E)$ for a left Haar measure λ ; G is unimodular when $\Delta = 1$.	Harmonic analysis	theorem folland_2_29_unimodular_of_compact_commutator_quotient {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] (hcpt : CompactSpace (G (commutator G).topologicalClosure)) (x : G) : Measure.modularCharacterFun x = 1 := by sorry
2.31 Theorem. If λ is a left Haar measure on G and Δ is the modular function of G , then for every f in $L^1(G)$ $\int_G f(x^{-1})\Delta(x^{-1}) d\lambda(x) = \int_G f(x) d\lambda(x)$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right translates of f , and $f^*g(x) = \int f(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. The modular function $\Delta : G \rightarrow (0,\infty)$ is determined by $\lambda(\Delta x) = \Delta(x)\lambda(E)$ for a left Haar measure λ ; it is a continuous homomorphism and is identically 1 exactly when G is unimodular.	Harmonic analysis	theorem folland_2_31_modular_inversion_formula {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] (f : $\mathbb{R} \rightarrow \mathbb{C}$) (hf : Integrable f mu) : integral x, f x^{-1} * ((Measure.modularCharacterFun x : $\mathbb{R} \Rightarrow \mathbb{R}$) : $\mathbb{R}$) dmu = integral x, f x dmu := by sorry
2.40 Proposition. Suppose $1 \leq p \leq \infty$, f in $L^p(G)$, and g in $L^p(G)$. (a) The integrals in (2.36) converge absolutely for almost every x , and we have f^*g in $L^p(G)$ and $\ f^*g\ _p \leq \ f\ _1 \ g\ _p$. (b) If G is unimodular, the same conclusions hold with f^*g replaced by g^*f . (c) If G is not unimodular, we still have g^*f in $L^p(G)$ when f has compact support. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right translates of f , and $f^*g(x) = \int f(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. The modular function $\Delta : G \rightarrow (0,\infty)$ is determined by $\lambda(\Delta x) = \Delta(x)\lambda(E)$ for a left Haar measure λ ; G is unimodular when $\Delta = 1$.	Harmonic analysis	noncomputable def groupConv (mu : Measure G) (f g : $G \rightarrow \mathbb{C}$) : $G \rightarrow \mathbb{C} :=$ fun x => integral y, f y * g (y^{-1} * x) dmu theorem folland_2_40_convolution_lp_bound {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] (p : $\mathbb{R} \Rightarrow \infty$) (hp : $1 \leq p$) (f g : $G \rightarrow \mathbb{C}$) (hf : Integrable f mu) (hg : MemLp g p mu) : ((forall^m x dmu, Integrable (fun y => f y * g (y^{-1} * x)) mu) /\ MemLp (groupConv mu f g) p mu /\ eLpNorm (groupConv mu f g) p mu <= eLpNorm f 1 mu * eLpNorm g p mu) /\ ((forall y : G, Measure.modularCharacterFun y = 1) -> MemLp (groupConv mu g f) p mu /\ eLpNorm (groupConv mu g f) p mu <= eLpNorm f 1 mu * eLpNorm g p mu) /\ (HasCompactSupport f -> MemLp (groupConv mu g f) p mu) := by sorry
2.42 Proposition. If $1 \leq p < \infty$ and f in $L^p(G)$ then $\ L_y f - f\ _p$ and $\ R_y f - f\ _p$ tend to zero as $y \rightarrow 1$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right translates of f , and $f^*g(x) = \int f(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. The modular function $\Delta : G \rightarrow (0,\infty)$ is determined by $\lambda(\Delta x) = \Delta(x)\lambda(E)$ for a left Haar measure λ ; G is unimodular when $\Delta = 1$.	Harmonic analysis	def leftTranslate (y : G) (f : $G \rightarrow \mathbb{C}$) : $G \rightarrow \mathbb{C} :=$ fun x => f (y^{-1} * x) def rightTranslate (y : G) (f : $G \rightarrow \mathbb{C}$) : $G \rightarrow \mathbb{C} :=$ fun x => f (x * y) theorem folland_2_42_translation_continuity_lp {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] (p : $\mathbb{R} \Rightarrow \infty$) (hp : $1 \leq p$) (hp' : $p \neq \infty$) (f : $G \rightarrow \mathbb{C}$) (hf : MemLp f p mu) : Tendsto (fun y => eLpNorm (leftTranslate y f - f) p mu) (nhds 1) (nhds 0) /\ Tendsto (fun y => eLpNorm (rightTranslate y f - f) p mu) (nhds 1) (nhds 0) := by

Natural language statement

Domain

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2.44 Proposition. Let U be a neighborhood base at 1 in G. For each U in U, let ψ_U be a function such that (i) $\text{supp } \psi_U$ is compact and contained in U, (ii) $\psi_U \geq 0$ and $\int \psi_U = 1$. Then $\ \psi_U * f - f\ _p \rightarrow 0$ as $U \rightarrow \{1\}$ if $1 \leq p < \infty$ and f in L^p, or if $p = \infty$ and f is left uniformly continuous. If, in addition, (iii) $\psi_U(x^{-1}) = \psi_U(x)$ for all x, then $\ f * \psi_U - f\ _p \rightarrow 0$ as $U \rightarrow \{1\}$ if $1 \leq p < \infty$ and f in L^p, or if $p = \infty$ and f is right uniformly continuous. A family $\{\psi_U\}$ satisfying (i)-(iii) is called an approximate identity. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right translates of f, and $f^*g(x) = \int f(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. The modular function $\Delta : G \rightarrow (0, \infty)$ is determined by $\lambda(\Delta x) = \Delta(x)\lambda(E)$ for a left Haar measure λ; G is unimodular when $\Delta = 1$.	Harmonic analysis	<pre> noncomputable def groupConv (mu : Measure G) (f g : G -> C) : G -> C := fun x => integral y, f y * g (y^-1 * x) dmu theorem folland_2_44_approximate_identity {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] (p : R>=0inf) (hp : 1 <= p) (hp' : p != inf) (f : G -> C) (hf : MemLp f p mu) (eps : R>=0inf) (heps : 0 < eps) : exists U in nhds (1 : G), forall psi : G -> R, HasCompactSupport psi -> tsupport psi subset U -> (forall x, 0 <= psi x) -> Integrable psi mu -> integral x, psi x dmu = 1 -> eLpNorm (groupConv mu (fun x => (psi x : C)) f - f) p mu < eps /\ ((forall x, psi x^-1 = psi x) -> eLpNorm (groupConv mu f (fun x => (psi x : C)) - f) p mu < eps) := by sorry </pre>
2.45 Theorem. Let I be a closed subspace of $L^1(G)$. Then I is a left ideal if and only if it is closed under left translations, and I is a right ideal if and only if it is closed under right translations. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right translates of f, and $f^*g(x) = \int f(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution.	Harmonic analysis	<pre> def leftTranslate (y : G) (f : G -> C) : G -> C := fun x => f (y^-1 * x) def rightTranslate (y : G) (f : G -> C) : G -> C := fun x => f (x * y) noncomputable def groupConv (mu : Measure G) (f g : G -> C) : G -> C := fun x => integral y, f y * g (y^-1 * x) dmu def IsLpClosed (p : R>=0inf) (mu : Measure G) (I : Set (G -> C)) : Prop := forall f : G -> C, MemLp f p mu -> (forall eps : R>=0inf, 0 < eps -> exists g in I, MemLp g p mu /\ eLpNorm (f - g) p mu < eps) -> f in I theorem folland_2_45_closed_ideals_are_translation_invariant {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] (I : Submodule C (G -> C)) (hclosed : IsLpClosed 1 mu (I : Set (G -> C))) (hmem : forall f in I, Integrable f mu) : ((forall g : G -> C, Integrable g mu -> forall f in I, groupConv mu g f in I) <-> forall (y : G), forall f in I, leftTranslate y f in I) /\ ((forall g : G -> C, Integrable g mu -> forall f in I, groupConv mu f g in I) <-> forall (y : G), forall f in I, rightTranslate y f in I) := by sorry </pre>
2.51 Theorem. Suppose G is a locally compact group and H is a closed subgroup. There is a G-invariant Radon measure μ on G/H if and only if $\Delta_G _H = \Delta_H$. In this case, μ is unique up to a constant factor, and if this factor is suitably chosen we have $\int_G f(x) dx = \int_{G/H} \int_H f(xh) d\mu(xh)$ for f in $C_c(G)$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right translates of f, and $f^*g(x) = \int f(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. $Pf(xH) = \int_G f(xg) dg$ is the averaging map $C_c(G) \rightarrow C_c(G/H)$, and Δ_G, Δ_H are the modular functions of G and of H.	Harmonic analysis	<pre> theorem folland_2_51_invariant_measure_on_quotient {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] (H : Subgroup G) (hH : IsClosed (H : Set G)) [LocallyCompactSpace H] (nu : Measure H) [nu.IsHaarMeasure] [TopologicalSpace (G / H)] [MeasurableSpace (G / H)] [BorelSpace (G / H)] : (exists rho : Measure (G / H), rho != 0 /\ (forall g : G, rho.map (fun q => g * q) = rho) /\ forall f : G -> C, Continuous f -> HasCompactSupport f -> integral x, f x dmu = integral q : G / H, (integral y : H, f (Quotient.out q * y) dnu) drho) <-> forall y : H, ((Measure.modularCharacterFun (y : G) : R>=0) : R) = ((Measure.modularCharacterFun y : R>=0) : R) := by sorry </pre>
2.69 Theorem. On any locally compact group G we have $L^1(G) L^p(G) = L^p(G)$ for $1 \leq p < \infty$. Moreover, $L^1(G)^* L^\infty(G) = L^\infty(G)^* C_{\{1\}}(G) = C_{\{1\}}(G)$ and $L^\infty(G)^* L^1(G) = C_{\{1\}}(G)^* L^1(G) = C_{\{1\}}(G)$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_y f(x) = f(y^{-1}x)$ and $R_y f(x) = f(xy)$ are the left and right	Harmonic analysis	<pre> noncomputable def groupConv (mu : Measure G) (f g : G -> C) : G -> C := fun x => integral y, f y * g (y^-1 * x) dmu theorem folland_2_69_convolution_factorization {G : Type*} [Group G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] </pre>

Natural language statement

Domain

GroundTruth

translates of f, and $f^*g(x) = \int g(y)g(y^{-1}x) dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. $C_{\{l\}}(G)$ and $C_{\{r\}}(G)$ denote the bounded left- and right-uniformly continuous functions on G.		<pre>(mu : Measure G) [mu.IsHaarMeasure] (p : ℝ>=0inf) (hp : 1 <= p) (hp' : p != inf) (f : G -> C) (hf : MemLp f p mu) : exists g h : G -> C, Integrable g mu /\ MemLp h p mu /\ forall^m x dmu, groupConv mu g h x = f x := by sorry</pre>
4.81 Theorem. If f is a bounded continuous function on G, the following are equivalent: a. f is the restriction to G of a continuous function on bG. b. f is the uniform limit of linear combinations of characters on G. c. f is uniformly almost periodic. Notation: G is a locally compact abelian group, G its dual group of continuous characters $\chi : G \rightarrow \mathbb{T}$, and $f(\chi) = \int f(x)\text{conj}(\langle x, \chi \rangle) dx$ the Fourier transform. For a closed ideal I subset $L^1(G)$, $\text{nu}(I) = \{\chi : f(\chi) = 0 \text{ for all } f \text{ in } I\}$ is its cospectrum (hull) and, for E subset G, $\text{iota}(E) = \{f \text{ in } L^1(G) : \int_E f = 0\}$ is the kernel of E; $\text{nu}(f) := \text{nu}(\{f\})$. bG is the Bohr compactification of G, and f is *uniformly almost periodic* when the set of its right translates $\{R_y f : y \text{ in } G\}$ is totally bounded in the uniform norm.	Harmonic analysis	<pre>def rightTranslate (y : G) (f : G -> C) : G -> C := fun x => f (x * y) def IsUniformlyAlmostPeriodic (f : G -> C) : Prop := forall eps : ℝ, 0 < eps -> exists s : Finset G, forall y : G, exists z in s, forall x : G, rightTranslate y f x - rightTranslate z f x < eps theorem folland_4_81_almost_periodic_characterization {G : Type*} [CommGroup G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] (f : G -> C) (hf : Continuous f) (hbdd : exists C : ℝ, forall x, f x <= C) : (forall eps : ℝ, 0 < eps -> exists (s : Finset (PontryaginDual G)) (c : PontryaginDual G -> C), forall x, f x - sum xi in s, c xi * (xi x : C) < eps) <-> IsUniformlyAlmostPeriodic f := by sorry</pre>